MIL-C-83398 (USAF)

7 November 1972

MILITARY SPECIFICATION

CLOTH, COATED, STRETCH, POLYCHLOROPRENE, KNITTED NYLON

1. SCOPE

1.1 Scope. This specification covers one type of polychloroprene-coated, one side only, nylon knit stretch cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-1136	Packaging and Packing of Coated (Plastic; Rubber: and Laminated Fabrics

Military

MIL-A-5540	Adhesive, Polychloroprene

STANDARDS

Federal

Fed. Std. No. 191	Textile Test Methods
Fed. Std. No. 601	Rubber: Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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3. REQUIREMENTS

3.1 First article sample. Before production is commenced, a 1 yard long, full width of the coated cloth, a 1/4 yard, full width of the knitted nylon base cloth, an adhesion of coating sample as described in 4.3.1.1 and sheets of cured coating compound which shall be 0.07 to 0.08 inch thick and 2 square feet in area, as specified in the contract or order shall be submitted. Each sample shall be accompanied by the contractor's test report and shall be submitted to the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the first article sample authorizes the commencement of production but does not relieve the supplier of the responsibility for compliance with all the provisions of this specification. The first article sample shall be manufactured by the supplier in the same facilities to be used for manufacture of the production items.

3.2 Materials. The finished coated cloth shall consist of knitted nylon base cloths uniformly coated with a polychloroprene compound.

3.2.1 Base cloth. The base cloth shall be a natural color, plain jersey circular knit, 2.2 - 2.5 ozs./sq. yd., made of nylon stretch yarns. The yarns shall be nylon 6-6 made from hexamethylene diamine and adipic acid. The color of the base cloth shall be natural.

3.2.2 Coating. The coating compound tested as specified (see 4.6.5) shall contain not less than 60 percent by volume of polychloroprene. The remainder of the compound shall consist of softeners, curing agents, anti-oxidants, and reinforcing materials. The coating shall be the natural color of polychloroprene or suitably pigmented during compounding so that the cured compound and base cloth shall have a matching color. The coating compound shall be compatible with the base cloth and contain no waxes or other ingredients that may bloom to the surface to adversely affect the coating adhesion, and cementability of the finished cloth. The ingredients of the compound shall be water insoluble after curing. The coating compound, when cured in sheet form to the same degree of cure and in the same manner as the finished product, shall have the physical properties specified in table I.

Table I. Physical properties - Coating compound

CHARACTERISTIC	REQUIREMENT
Tensile strength:	
Original, psi, min.	1300
After 96 hrs., accelerated aging, % loss, max.	10
After 100 hrs., accelerated weathering, % loss, max.	20
Elongation:	
Original, %, min.	500
After 96 hrs., accelerated aging, % loss, max.	10
After 100 hrs., accelerated weathering, % loss, max.	20

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3.3 Application of coating. After application of a foundation coat, which must be compatible with the base cloth and the remainder of the coating compound to produce the adhesion specified, the coating compound shall be applied by a spreader or calender coating operation on the side showing lengthwise ribs, crosswise courses.

3.4 Dusting powder. The cured polychloroprene coating may be lightly dusted. The dusting powder for use on the chloroprene coating shall be whiting, talc, zinc stearate or other finely divided mineral matter which does not support mildew growth (see 4.4).

3.5 Physical properties. The physical properties of the coated cloth shall conform to the requirements specified in table II and in the following subparagraphs when tested as specified therein.

Table II. Coated cloth requirements

CHARACTERISTIC	REQUIREMENT
Weight, oz./sq. yd., max.	13.0
Breaking strength, 2-inch wide strip, pounds, min.	
Wales - (In direction of)	55
Courses - (In direction of)	20
Elongation at break, 2-inch wide strip, percent, min.	
Wales - (In direction of)	100
Courses - (In direction of)	300

3.5.1 Coating thickness. The coating on the cloth shall be not less than 0.007 inch or more than 0.0105 inch thick (see 4.6.6).

3.5.2 Adhesion. The coating adhesion to the base cloth shall be not less than 8 pounds for a 2-inch wide specimen when tested in the course direction (see 4.6.7).

3.5.3 Modulus at 75 percent elongation. The load required to maintain 75 percent elongation on a 2-inch wide specimen, after the specimen has previously been stretched 100 percent, shall not exceed 15.0 pounds in the wales direction and 5 pounds in the course direction (see 4.6.1).

3.5.4 Resistance to flexing. The coated cloth shall be subjected to 1000 flexing cycles at 50 percent stretch in the course direction. After flexing, the specimen shall have no tears in the coating, no separation of coating from the cloth, nor shall it leak water (see 4.6.2).

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3.5.5 Resistance to low temperature exposure. After low temperature exposure for 30 minutes at -40° ± 1.1° C (-40° ± 2° F) the specimen shall show no signs of cracking, and it shall not leak water (see 4.6.3).

3.5.6 Tension set. After being stretched to 100 percent for 16 hours, the tension set on the specimens shall not exceed 20.0 percent when examined 3 hours later (see 4.6.4).

3.6 Dimensions

3.6.1 Length. The coated cloth shall be furnished in rolls. The minimum length of a roll shall be 100 yards, the maximum number of pieces in a roll shall be 4 and the minimum length of the shortest piece in a roll shall be 10 yards.

3.6.2 Width. The coated cloth shall be furnished in the width specified by the ordering data (see 6.2).

3.7 Workmanship. The coated cloth shall be uniformly processed to produce a high quality product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs) specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection

4.2.1 Sampling for inspection. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein. Quality conformance tests are required on each lot (see 4.2.2) of material furnished to this specification.

4.2.2 Lot. A lot shall consist of all material of the same identity, cured in the same production run, and submitted at the same time for inspection.

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4.2.3 Inspection of materials and components. The supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification (see 4.1). In the event of conflict, this specification shall govern.

4.2.4 Yard-by-yard examination. The required yardage of each roll shall be inspected on both sides and visual defects classified as listed. In addition, the coated side shall be given a separate through-light inspection for pinholes, light areas and windows. The through-light inspection shall be performed in a darkened area using the lighting conditions described below.* During the examination when the surface of the coated cloth is in contact with the light table, the illumination in the darkened room shall not exceed 25 foot candles of natural or artificial light. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case, only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 yard. The acceptable quality level (AQL) shall be 10.0 defects per hundred units (yards). The inspection level shall be Level II of MIL-STD-105. The number of rolls over which the sample yardage shall be distributed shall be in accordance with table III. An approximate equal number of yards shall be examined from each roll selected.

*Light table. The light table shall have a clear glass top and shall be illuminated with a minimum of two 25 watt fluorescent tubes. The tubes shall be positioned 9 to 10 inches below the glass top and 6 to 8 inches from the sides and ends of the light housing. The spacing between tubes shall be 5 to 6 inches and the interior of the light housing shall be a flat white color.

Part A - Coated side:

Defect
Any cut, hole, tear or abrasion
Any pinhole
Any uncoated area
Any scorch or burn
Any blister, tunnel or delamination of coating
Splice mark clearly visible on coated side
Any lump or heavily coated area

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Crease or wrinkle - Resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure
Uneven coating - thin area where coating is missing or noticeably thinner
Slub, slug, kink or knot resulting in coating streak or uneven coating
Any spot, stain, or streak over 1 inch in length, any direction, (clearly visible at approximately 3 feet)
Any foreign matter which on removal leaves an uncoated or poorly coated area
Width less than minimum specified
Any objectionable odor (not characteristic of coating compound)
Ripples, waviness, dimensional distortion
Edges not cut straight, folded or rolled
Variation in color or coating
Tackiness (coating will adhere and not readily unroll)
Color does not match standard or specified sample

Part B - Uncoated side:

Defect

Any hole
Dropped stitch or run
Snag or pull affecting coating

Table III. Sample size

LOT SIZE IN YARDS	SAMPLE SIZE IN ROLLS	ACCEPTANCE NO.
Up to 1200 1/ inclusive	3	0
1201 to 3200 inclusive	5	0
3201 to 10,000 inclusive	8	0
10,001 to 35,000 inclusive	13	0
35,001 to 150,000 inclusive	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls each roll in the lot shall be examined.

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4.2.5 Examination for length

4.2.5.1 Individual roll length. The unit of product shall be one roll. The number of rolls to be examined and the acceptance number shall be in accordance with table II. The following shall be considered a defect:

Any roll not having the coated side of the fabric facing as specified
Any roll (gross length) less than minimum or more than maximum specified
Any roll (gross length) more than 2 yards less than gross length marked on ticket
Any splices or seam in any roll (if not allowed)
Any roll containing more than specified number of pieces
Any piece in roll less than minimum specified length
Any piece in roll less than minimum specified width.

4.2.5.2 Total yardage in samples. The rolls examined shall be those selected for examination of individual rolls as specified in 4.2.4. The lot shall be unacceptable if the total of the gross lengths of the rolls in the sample is less than the total of the gross lengths marked on the tickets.

4.3 First article inspection. First article samples shall be subjected to all tests and inspections set forth in this and component specifications to the extent applicable to determine materials conform to established requirements.

4.3.1 Adhesion of coating samples

4.3.1.1 Manufacturer's prepared samples. The manufacturer's samples shall be prepared by the manufacturer (coater). Two pieces measuring 12 by 12 inches shall be taken from the coated cloth prior to curing. The two pieces shall be placed together, coating to coating, except for a 1 inch wide separation along a filling edge allowed by the insertion of paper. The specimen shall be cured in the same manner and degree of cure as the finished product and tested as specified in 4.6.

4.3.1.2 Adhesive prepared sample. If the manufacturer's prepared samples are not available, the sample shall be prepared using adhesive conforming to MIL-A-5540 and tested as specified in 4.6.

4.4 Certificate of compliance. A certificate of compliance shall be furnished with each shipment or lot stating the following requirements have been met:

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a. The compound in the coating contains a minimum of 60 percent polychloroprene, by volume (see 3.2.2).

b. The remainder of the coating compound consists of softeners, curing agents, anti-oxidants and reinforcing materials (see 3.2.2).

c. The coating compound contains no waxes or other water soluble ingredients (see 3.2.2).

d. The dusting powder does not support mildew growth (see 3.4).

e. Coating compounds will not promote skin irritation or have detrimental effect on nylon.

4.5 Quality conformance tests. Quality conformance tests shall consist of all the tests specified or as many of the tests deemed necessary to assure that the production material meets all the requirements specified herein.

4.5.1 Rejection criteria. Failure to meet the test requirements specified herein shall constitute cause for rejection of the lot.

4.6 Testing. The methods of testing specified in Fed. Std. No. 191 or Fed. Std. No. 601 wherever applicable and as listed in table IV shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The sample unit for testing shall be:

a. One continuous yard, full width of the coated cloth.

b. Two coating adhesion test specimens (see 4.2.1) in course direction.

c. Two square feet in area of cured coating compound, 0.07 to 0.08 inch thick.

d. One fourth yard full width of the knitted base cloth.

The lot size shall be expressed in units of 1 linear yard and the sample size (number of sample units) shall be as specified below. The lot shall be unacceptable if any unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

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Table IV. Test methods

ITEM	TEST CHARACTERISTIC	REQUIREMENT PARAGRAPH	TEST METHOD
Coating Compound	Polychloroprene	3.2.2	4.6.5
	Tensile Strength, initial	Table I	4.6.8
	After accelerated aging	Table I	4.6.8/4.6.9
	After accelerated weathering	Table I	4.6.8/4.6.10
	Elongation, initial	Table I	4.6.8
	After accelerated aging	Table I	4.6.8/4.6.9
	After accelrated weathering	Table I	4.6.8/4.6.10
Base Cloth	Weight	3.2.1	5041
Coated Cloth	Weight 1/	Table II	5041
	Breaking Strength & Elongation	Table II	5102
	Thickness of Coating	3.5.1	4.6.6
	Coating Adhesion	3.5.2	4.6.7
	Modulus	3.5.3	4.6.1
	Resistance to Flexing	3.5.4	4.6.2
	Resistance to Low Temperature	3.5.5	4.6.3
	Tension Set	3.5.6	4.6.4

1/ Breaking Strength and Elongation shall be obtained simultaneously on the same specimen.

4.6.1 Modulus. Test specimens, 2 ± 0 by 3 ± 1/2 inches, shall have a 1 inch gage length inscribed and shall be tested in a low capacity tester 50 pound maximum), operated at a speed of 2 inches per minute. The tester shall be operated to stretch the 1 inch gage marks to 2 inches and then immediately reversed to decrease the gage length to 1-3/4 inches. The number of pounds indicated on the test machine at the 75 percent elongation shall be recorded as the modulus.

4.6.2 Resistance to flexing. A test specimen 8 inches wide and at least 8 inches long in the course direction, shall be marked with a 6 inch gage length in the center of the course direction. The length of the course direction shall be governed by the clamps of the flexing machine. After the specimen has been subjected to the specified flexing cycles at a speed of 10 cycles per minute, the specimen shall be examined for evidence of tears in the coating and separation of the coating from the cloth. The specimen shall then be tested for water leaks in accordance with Method 5516 of Fed. Std. No. 191. The water shall be applied for 3 hours at a height of 60 inches.

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4.6.3 Resistance to low temperature exposure. Resistance to low temperature exposure shall be determined in accordance with Method 5874 of Fed. Std. No. 191. The specimen shall be subjected to a temperature of $-40^{\circ} \pm 1.1^{\circ}$ C ($-40^{\circ} \pm 2^{\circ}$ F) for a total of 30 minutes. The specimen shall be examined for evidence of cracking. The specimen shall then be tested for water leaks in accordance with Method 5516 of Fed. Std. No. 191. The water shall be applied for 5 minutes at a height of 60 inches.

4.6.4 Tension set. A test specimen, identical with the modulus specimens of 4.6.1, shall be maintained at 100 percent elongation in the course direction for 16 hours. The elongation shall be released at a rate of 0.50 inches per minute. After allowing the specimen to recover for 3 hours, the gage length shall be measured and the percent tension set determined.

4.6.5 Polychloroprene compound. The volume of polychloroprene present in the coating compound shall be determined in accordance with Method 15111 of Fed. Std. No. 601, the following conversion formula for percent of polychloroprene by volume:

$$\text{Percent} = \frac{\text{percent by weight of polychloroprene X specific gravity of sample}}{\text{specific gravity of polychloroprene } 1/}$$

1/ Specific gravity of polychloroprene is 1.27.

4.6.6 The coating thickness shall be determined in accordance with Method 2011 of Fed. Std. No. 601, except that the thickness shall be determined using the coated cloth thickness minus the knitted base cloth thickness.

4.6.7 The coating adhesion test specimen shall be subjected to the adhesion test in accordance with Method 8011 of Fed. Std. No. 601.

4.6.8 Tensile strength and elongation of coating compound. The tensile strength of cured coating compound samples shall be determined before and after exposure to the specified accelerated aging and accelerated weathering tests. Tensile strength and ultimate elongation shall be determined in accordance with Methods 4111 and 4121 of Fed. Std. No. 601. The Type III die shall be used.

4.6.9 Accelerated aging of coating compound. The cured coating compound samples shall be subjected to 96 hours of accelerated aging in accordance with Oxygen Pressure Test Method 7111 of Fed. Std. No. 601.

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4.6.10 Accelerated weathering of coating compound. The cured coating compound samples shall be tested in accordance with Method 7311 of Fed. Std. No. 601. All specimens shall be stretched to 10 percent elongation. The stretched samples shall be exposed for 100 hours in the weathering unit.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Cloth shall be packaged, packed, and marked in accordance with PPP-P-1136, except that, prior to rolling, the face of the coated cloth shall be dusted with a suitable talc to prevent tackiness and sticking. The levels of packaging and packing shall be as specified by the procuring activity (see 6.2). In addition to the markings required, each tag shall be marked with the following information: "Date of Application of Coating". In addition, the following precautionary marking shall appear on all containers: "STORE IN A COOL DRY PLACE".

6. NOTES

6.1 Intended use. The cloth is intended for use in the fabrication of protection clothing.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity and width.

c. Selection of applicable levels of packaging and packing (see 5.1).

d. First article.

6.3 Instrument for measuring light intensity. The Weston Illumination Meter Model 756 or 703, type 8 with visor filter made by Daystrom, Inc., Weston Industries, Newark, New Jersey.

6.4 First article sample. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article material shall be those material as specified in 3.1.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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